Forensic Biometrics & Master Persona Audit

TOTAL CATALOGED: 1,652 Personas (SUBJ-01 to SUBJ-1652)
METHODOLOGY: InsightFace ArcFace 512-D Vectors | DBSCAN Cosine eps <= 0.38
GROUND TRUTH VERIFICATION: Operator SUBJ-1652 formally reconciled from manual review.

1. Tier 1 Core Plant Personnel (50+ Detections)

Subject ID	Total Dets	Active Days	First Seen	Last Seen	Contacts	Risk
SUBJ-01	29,598	29	2026-09-01 01:29	2026-09-30 23:32	**116**	**CRITICAL**
SUBJ-02	166	12	2026-09-01 09:04	2026-09-30 08:51	**53**	**CRITICAL**
SUBJ-03	104	2	2026-09-01 07:29	2026-09-11 07:46	**0**	**LOW**
SUBJ-04	87	16	2026-09-01 07:29	2026-09-30 08:13	**0**	**LOW**
SUBJ-05	86	2	2026-09-16 16:19	2026-09-23 11:21	**0**	**LOW**
SUBJ-06	85	12	2026-09-02 08:36	2026-09-30 09:14	**0**	**LOW**
SUBJ-07	81	3	2026-09-01 11:14	2026-09-21 11:22	**0**	**LOW**
SUBJ-08	79	10	2026-09-01 11:14	2026-09-29 11:14	**0**	**LOW**
SUBJ-09	79	1	2026-09-25 14:03	2026-09-25 14:15	**0**	**LOW**
SUBJ-10	75	2	2026-09-13 19:40	2026-09-21 19:37	**0**	**LOW**
SUBJ-11	69	9	2026-09-01 14:14	2026-09-30 09:13	**0**	**LOW**
SUBJ-12	68	2	2026-09-07 17:51	2026-09-09 17:50	**0**	**LOW**

2. Critical Equipment Protection Zone Breach Ledger

Exactly 17 individuals established direct physical contact (IoU > 0.08) with high-hazard plant assets:

Subject ID	Contacts	Active Days	Total Dets	Primary Equipment Asset	Hazard
SUBJ-01	**116**	29	29598	Two Multiport Backwash Valves	**CRITICAL**
SUBJ-02	**53**	12	166	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-102	**5**	1	8	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-167	**5**	1	6	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-41	**4**	10	20	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-344	**3**	1	3	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-395	**3**	1	3	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-129	**2**	1	7	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-489	**2**	1	2	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-22	**1**	1	35	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-342	**1**	1	3	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-582	**1**	1	2	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-759	**1**	1	2	Ultimate Plant Power Switch	**SEVERE**
SUBJ-842	**1**	1	2	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-890	**1**	1	2	Two Multiport Backwash Valves	**CRITICAL**
SUBJ-993	**1**	1	2	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-1079	**1**	1	2	Ultimate Plant Power Switch	**SEVERE**
SUBJ-1652	**1**	1	28	Two Multiport Backwash Valves	**CRITICAL**

3. Forensic Investigation: SUBJ-1652 vs SUBJ-02

FORENSIC GROUND TRUTH FINDING:
- **SUBJ-02:** Morning walkway operator at 09:04:32 (dark navy attire, exemplar face_0620).
- **SUBJ-1652:** Manual backwash valve operator at 09:53:38 (teal polo shirt, curly hair, mustache).
- **InsightFace 512-D Cosine Similarity: 0.1831 (NON-MATCH).** Proves two distinct individuals.